



COS10025 Technology in an Indigenous Context Project unit

School of Science, Computing, and Engineering Technologies

Seminar Week 5: Indigenous **STEM** Knowledge

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Acknowledgement of Country

We honor and celebrate the rich tapestry of Vietnam’s Indigenous peoples, encompassing 54 distinct ethnic groups, including the Kinh, Tay, Thai, Muong, Co Tu, Ede, Khmer, Hmong, Cham and many others, whose ancestral stewardship of this land spans millennia.

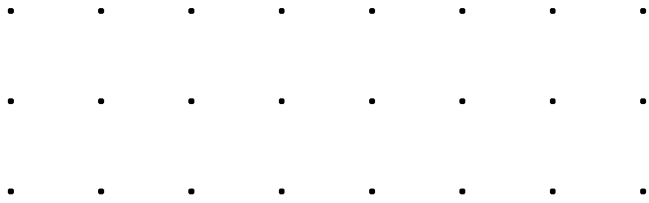
We pay our deepest respect to our ancestors past, present and future for preserving Vietnam’s cultures, history, timeless traditions and deep spiritual connection to its mountains, rivers, seas and forests.

We recognize their enduring heritage, resilience, and invaluable contributions to the nation’s diverse identity, committing to learn from and uplift their stories with reverence and gratitude.

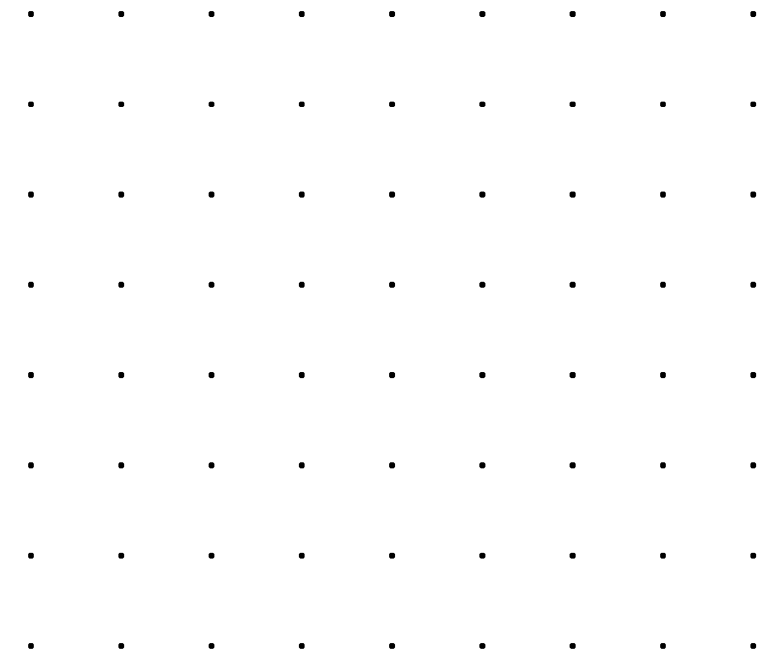


. . The worshipping of Hung Kings is a unique beauty in the spiritual culture of Vietnamese people . .

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Indigenous STEM Knowledge





RecapWeek 4

Indigenous Knowledges

We must remember, that Indigenous knowledges are specific to individual mobs, Indigenous groups across Australia are diverse, and have very different cultures, languages and customs. So too, are their health knowledges – particularly those around certain plants and animals which are specific to place.

However, the constant amongst all Indigenous mobs, is the holistic nature of Indigenous health and the inclusion of the “metaphysical, holistic, oral/symbolic, relational and intergenerational” (Levac et al. 2018)

Indigenous health is much broader than western models of health – as it includes interpersonal relationships, emotions, as well as physical and spiritual.

Even Indigenous understandings of learning and intelligence are intrinsically linked to nature and land, and the duality of the lived experiences of mob, plants and animals.



Western Knowledges

Western notions of intelligence differ strongly from Indigenous understandings – they ignore spirituality and connection, instead focusing on objectivity and “scientific”* rational.

The western model is much more anthropocentric, positioning humans at the top of the hierarchy of worldly order, and not plants, animals or the land.

However, Levac et al. (2018) states:

“there are schools of Western philosophical thought that lend themselves more to Indigenous ways of knowing, such as feminist and narrative theories that privilege storytelling”



Fire Farming

Indigenous communities used fire across Australia.

The burning, while reducing risk of uncontrollable fires, also created expansive grasslands, ensured good soils and reproduction of plants which encouraged kangaroos and other animals to come to the regenerated sections, where they could easily be hunted for food.

Selecting what areas to burn, when, with what method and how often, was part of place-specific Indigenous knowledges of the land.

After invasion, fire farming became less used – as the Europeans feared fire.

This meant that thick scrub and tall grasses grew in previously controlled and carefully managed areas, increasing the prevalence of bushfires.



Case Study in Vietnam: Mekong Delta faces droughts and saltwater intrusion

Vietnam's Mekong Delta is the third-largest delta on Earth and home to nearly 18 million people, whose livelihoods rely mainly on agricultural and aquacultural production. Considered the "Rice Bowl" of Vietnam, the Delta supports more than half of the country's total rice production and 95% of rice exports, contributing to Vietnam's position as the world's third-largest rice exporter. The region also produces 70% of the country's fruits and more than 60% of the country's seafood exports.

The region has recently faced the most severe periods of drought and saline water intrusion in the last 100 years—notably in 2015–2016 and even more devastatingly in 2019–2020. Dry season droughts have become increasingly severe year over year, forcing the Government to declare a state of emergency and issue a call for international support. Increased salinity hit 10 out of 13 provinces in the Mekong Delta during the 2019 and 2020 seasons, affecting 58,000 hectares (ha) of rice, 6,650 ha of fruit trees, 1,241 ha of vegetables, and 8,715 ha of aquaculture. Up to 96,000 households or 430,000 people faced a shortage of daily use water.²In addition, a thousand spots of subsidence have been recorded around the Delta, causing substantial damage to roads and houses.



<https://vietnam.opendevelopmentmekong.net/topics/droughts-and-saltwater-intrusion/>

Indigenous ways of knowing and health

Holistic health

Traditional healing and medicine

Caring for Country as necessary for holistic wellbeing

Sorry business

Men's and Women's business (gender specific Lore, custom and roles)

Health being a communal state, rather than individual



What is Indigenous Knowledge

Indigenous knowledges also referred to traditional knowledge incorporates the know-how, practices, skills and innovations. It can be found in a wide variety of contexts, such as agricultural, scientific, technical, ecological and medicinal fields. As well as biodiversity-related knowledge.

It is intertwined within cultural and social practices and the vast Indigenous languages



Relevance to science and innovation

Associate Professor Rowena Ball outlines examples of Indigenous scientific and engineering heritage, including the system of stone fish traps of the Ngunnhu people on the Barwon at Brewarrina; aquaculture and eel farming by the Gunditjmara people of south-western Victoria; and the work of eminent Indigenous scientist and inventor David Unaipon, ‘who extensively documented the scientific abilities and achievements of Indigenous peoples over Australia’.

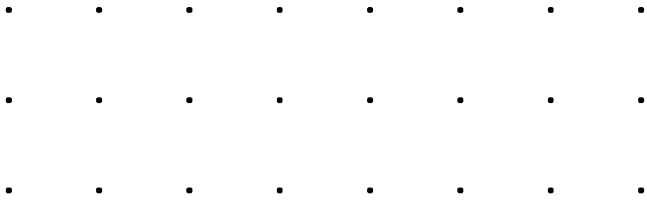
The 2013 report, [*Indigenous engagement with science: Towards deeper understandings*](#), noted that there are ‘significant opportunities for government and industry to engage with Indigenous people in a way that will maximise the potential for increased productivity across a wide range of scientific activity.’



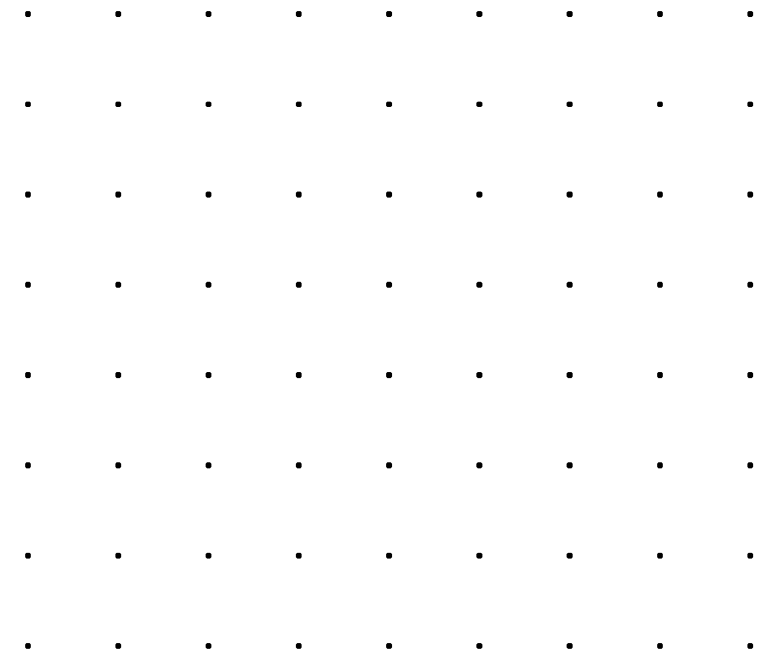
What will be cover today?

- Astronomy from the Indigenous context
- Use of nanofibers from spinifex and other resin trees
- Traditional use of Kakadu plum and how it is being utilised today (case study in Australia)
- Bamboo and Lotus in Vietnam, and how it is being utilised today (case study in Vietnam)





Astronomy & Narrative Science



Narrative Science

Storytelling is the way we transmit knowledges and educate our communities.

Story serves many purposes, and within our stories there are life lessons and rules for living, as well as detailed understanding of our countries, waterways, landscapes and seascapes and the world above us.

Aboriginal and Torres Strait Islander Australians view the sky as mirroring the earth.

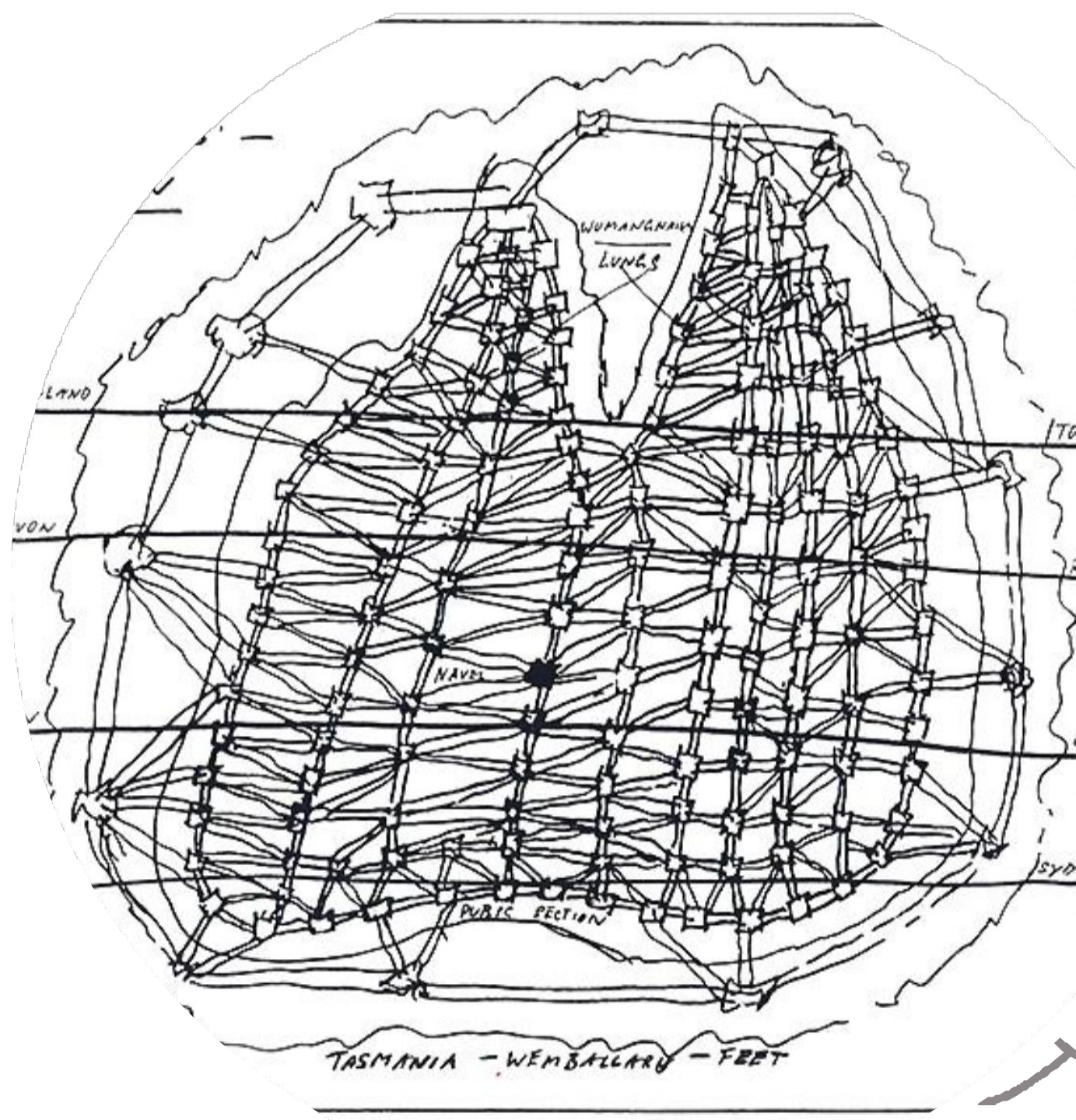
Our stories are more than beautiful tales they are narrative science and knowledge and rules for living.



Navigation & Astronomy



Songlines & Dreaming Tracks



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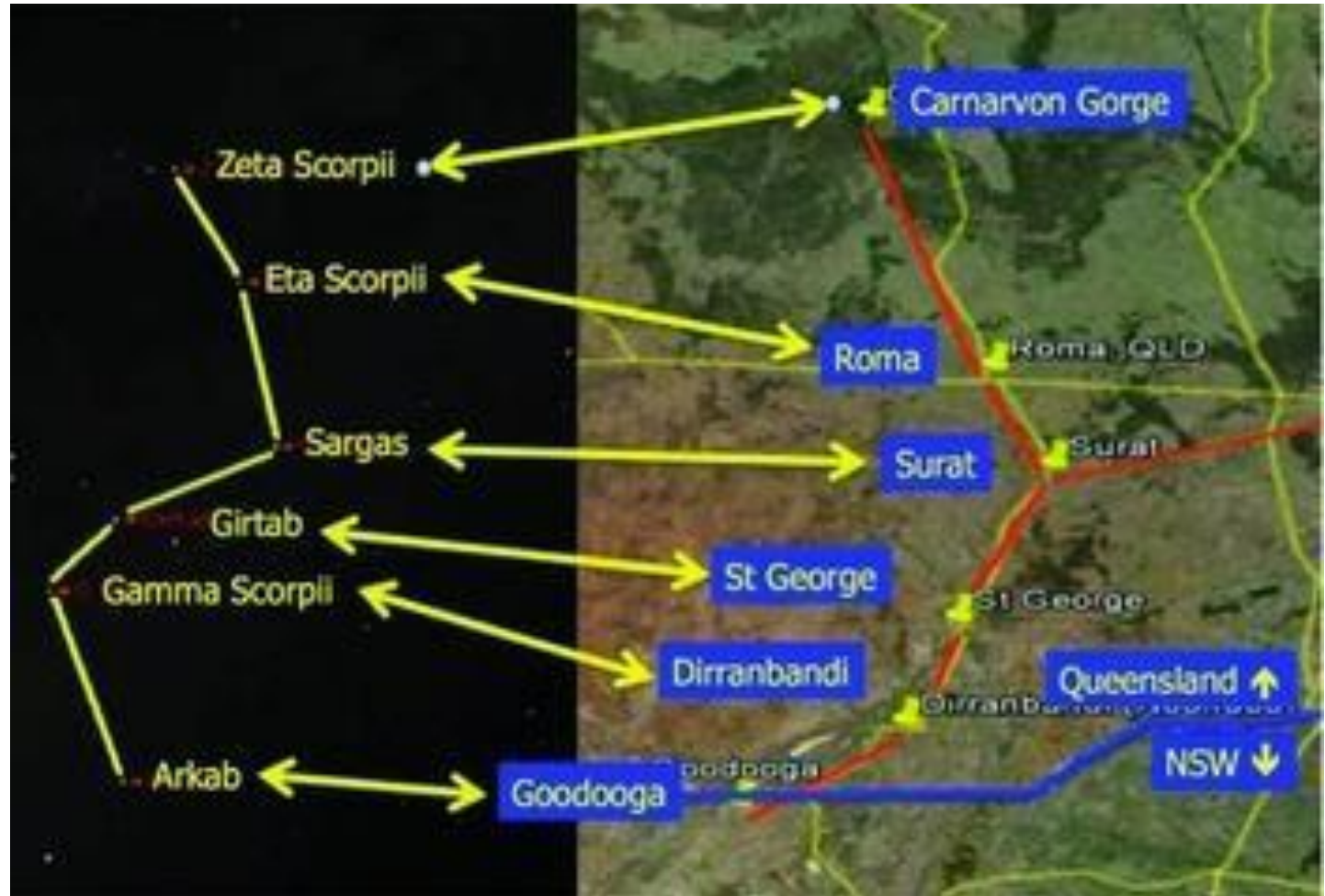


“Songlines are epic creation songs passed to present generations by a line of singers continuous since the dreamtime. These songs, or song-cycles, have various names according to which language group they belong to, and tell the story of the creation of the land, provide maps for the country, and hand down law as decreed by the creation heroes of the dreamtime. Some songlines describe a path crossing the entire Australian continent.”

(Wositsky & Harney 1999, p.301)

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Euahlayi
Songline:
between QLD
and NSW.



“The Dreaming Track in the sky! Planets making the pathway! Travelling routes, a pathway you could call it, like a highway! Travelling pathway joins to all different areas, to base place, to camping place, to ceremony place, where the trade routes come in; all this sort of things.

The Dreaming Track in the sky, the planets come straight across... walking trail becomes a pad, then becomes a wagon road, two wheel tracks, then

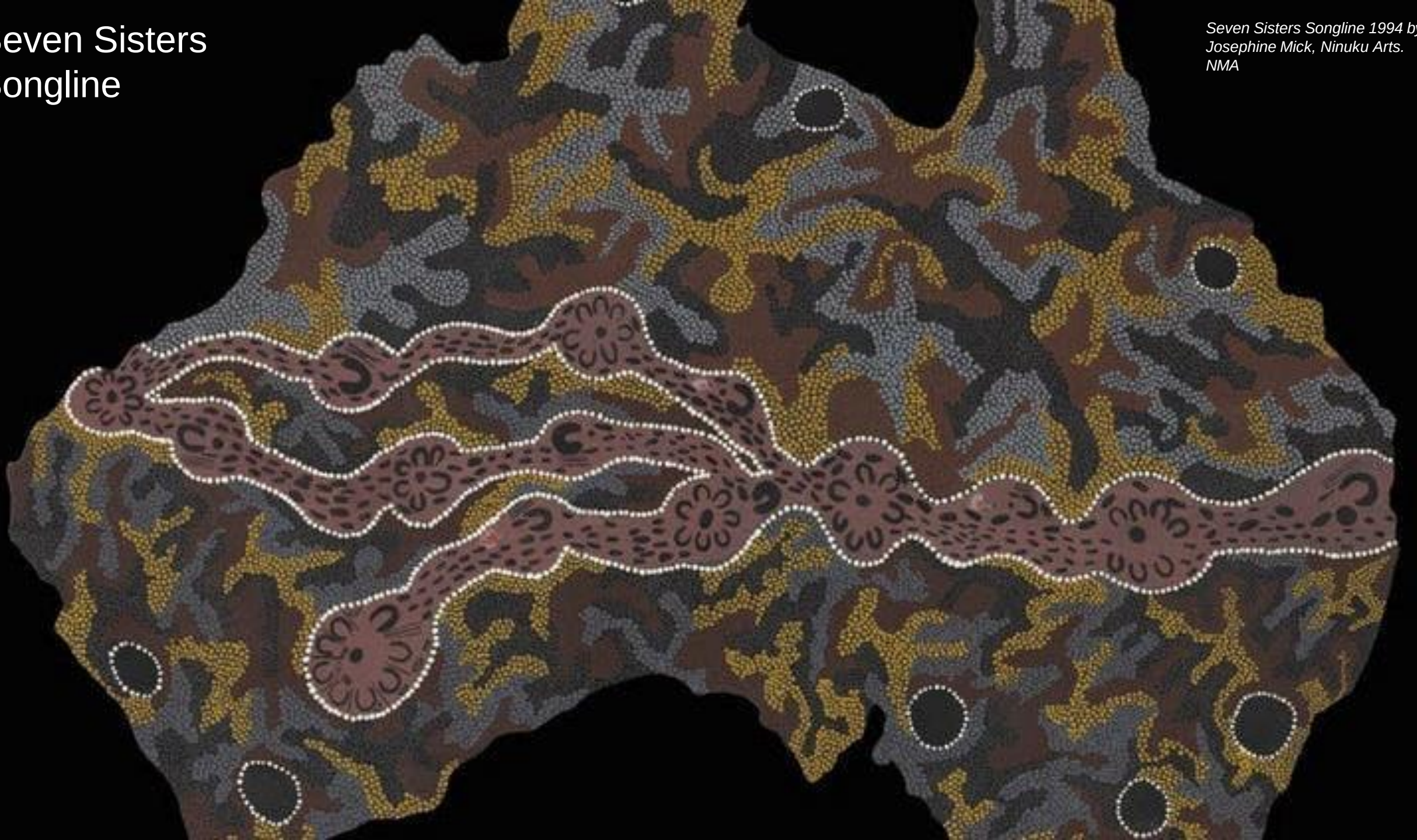
become a highway”

(Dawn Morris 2014).



Seven Sisters Songline

*Seven Sisters Songline 1994 by
Josephine Mick, Ninuku Arts.
NMA*



Seven Sisters

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Pleiades





Within Australia, the dreaming stories of the Seven Sisters are among the most common astronomy story told.





Star navigation & moon shadow travel



Songlines were used for long haul navigations:

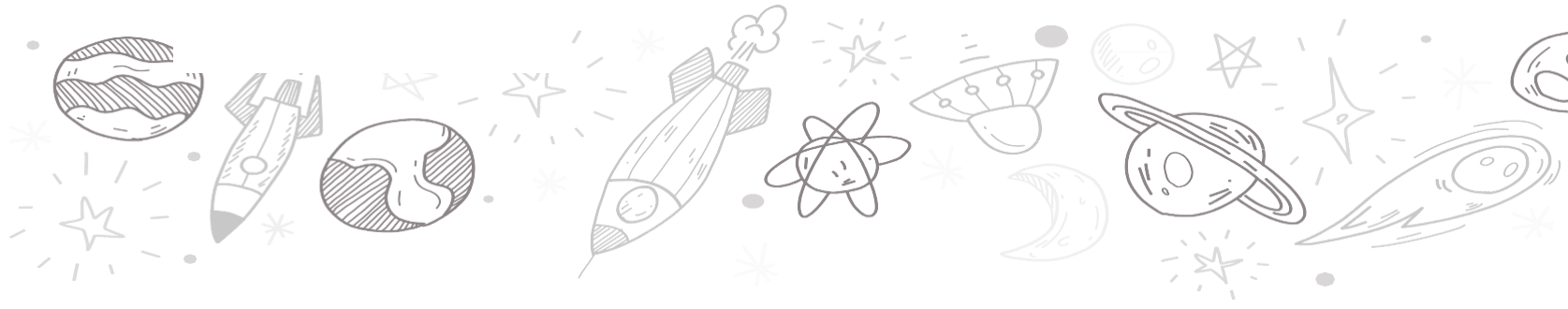
“Not just songline trail, walking trail, trade routes. You sing a song, then you follow your song, in that track you go along singing the song, like a blazed mark”

(Norris & Harney 2014).

When no songline existed, or for much shorter travel, mob would use the moon and stars alone for patterns and directions.



In summary



Indigenous stories are never just stories – but are complex knowledges, bound in culture and lore, which speak to the interconnection of the earth, the sky, plants, animals and waterways.

Slowly, modern Western science is looking to our peoples and our ancient knowledges and realising that Aboriginal and Torres Strait Islander Australians have had these knowledges well before the West, and that we might have knowledges of the sky and earth, that they do not yet possess.

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Nanofibres and spinifex resin

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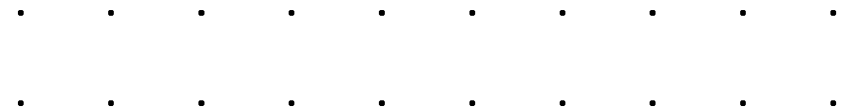
Case study: Spinifex

Scientists at the University of Queensland worked with the local Injаланj Didanu community in central Australia to identify and harvest specific spinifex species to extract nanofibers and resin. The project team will combine indigenous knowledge of Spinifex and its resin applications, manufacture, structure, and properties with controlled laboratory refining, modification, and testing to develop resins and fibres into renewable materials. We evaluated the possibility of developing as Principal investigator Professor Martin says the finding is unique stating:

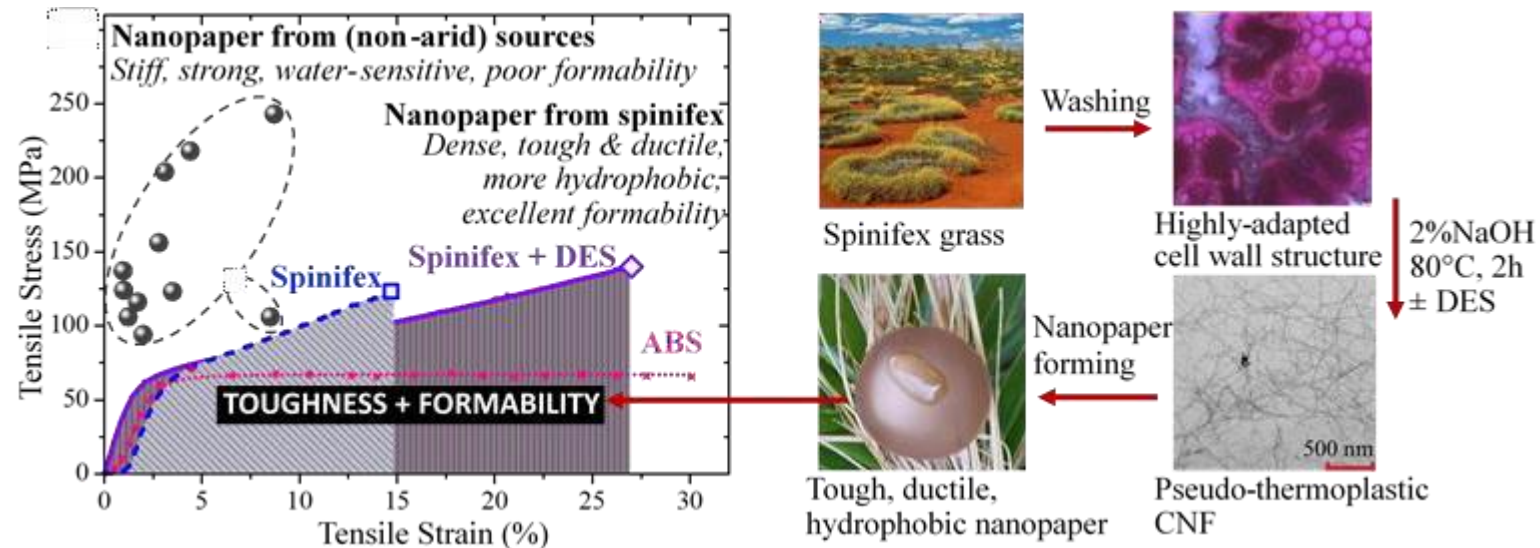
The nanofibres that we can extract are long, thin and stretchy – only a few nanometres wide but thousands of nanometres in length...As a materials scientist, this is exactly what we look for when we want to reinforce flexible materials...

The technology is expected to be of great interest to applied materials science industries, including the [multi-billion dollar condom market](#), and in carbon fibre and composites manufacturing (used for example, in aircraft, high-end cars and bikes).

This project has the potential to significantly improve prospects for employment and economic development for Indigenous people. Harvesting and initial processing of the spinifex will be undertaken in the community, with local rangers managing the environment where the spinifex grows. Colin Saltmere, Managing Director of the Dugalunji Aboriginal Corporation, said 'It's about providing jobs to our people and reclaiming some integrity'.



Formation of nano paper

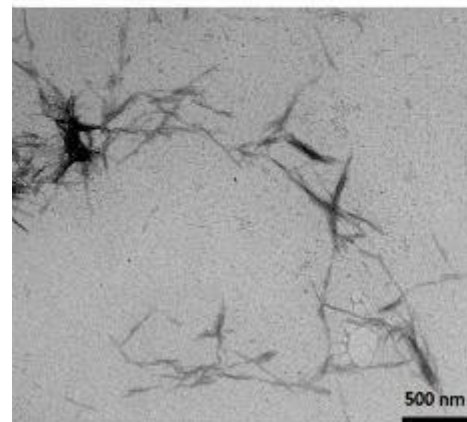


Producing nanofibres from spinifex

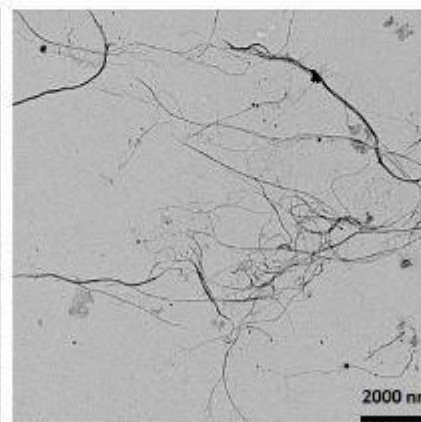
First, the grass is washed with hot water, dried and ground into a powder. Then, we isolate the nanofibres.

The grass powder is treated with a mild alkaline solution to loosen its structure before it is deconstructed using a high-pressure homogeniser, the same instrument used for homogenising milk.

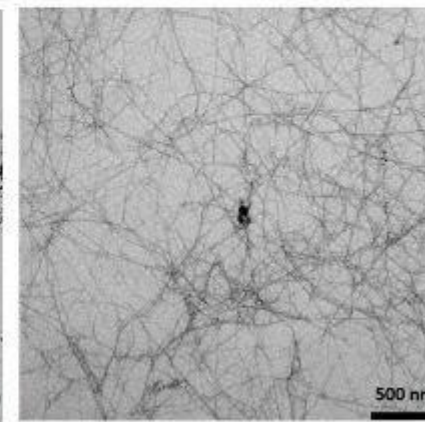
These nanofibres, either in powder form or dispersed in water, are then mixed with other materials such as rubbers, latex or cardboards to improve their strength while retaining flexibility and stretchiness.



Cellulose Nanocrystals from cotton



Cellulose Nanofiber from wood

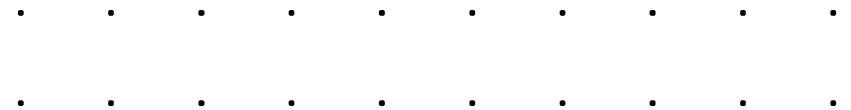


Cellulose Nanofiber from spinifex



Traditional use of spinifex

- Spinifex is a type of arid plant
- Grown in Australia for over 15 million years
- Traditionally used within the hafting process
- Some species produce a sticky resin, which Aboriginal people have used as an adhesive. But spinifex also has food, medicinal and architectural uses.



How spinifex can improve...

Cardboard and filters

Using spinifex nanofibre as a water filter significantly improves the removal of metal ions in water.

The hemicellulose on the surface of nanofibres works as absorption cage to trap metal ions.

Reinforcement of recycled cardboards with tough spinifex nanofibres improves its strength, which generally decreases in the recycling process.

Hemicellulose acts as an adhesive during the cardboard manufacturing process and imparts toughness and strength.



How spinifex can improve...

Elastomers and polymers

For stretchy, rubber-based products like gloves and condoms, the ideal material is strong and very flexible.

This combination will result in a glove or condom that is resistant to breakage

The common reinforcing agents used in rubbers to improve their mechanical properties are silica, carbon black, carbon nanotubes and graphene.

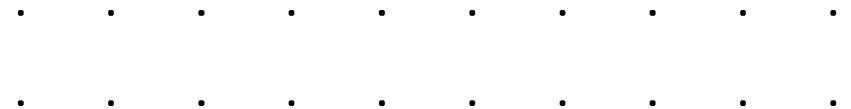
These can provide strength but, due to their rigidity, can also make the material too stiff or less stretchy.

A solution to this is using a soft and tough filler — such as spinifex nanofibres — to produce a thinner, tactile membrane.

In other rubber applications, such as seals, tyres and boots, similar technical requirements exist.

For all of these, toughness and abrasion resistance is key while still retaining the soft, elastic nature of rubber.

Adding just a small amount of spinifex nanofibres into different types of rubber results in a significant improvement in the strength of rubbers without reduction of their resilience and flexibility.



How spinifex can improve...

Concrete

Spinifex nanofibre has shown to increase the strength of cement by at least 20 per cent. This means that we can reduce the amount of cement needed or we can make thinner structures.



Case study in Vietnam: The Rise of Sustainable Bamboo Products

In recent years, there has been a significant increase in the demand for sustainable products. Consumers are becoming more conscious of the environmental impact of their purchasing decisions and are actively seeking out alternatives that are eco-friendly. One such alternative that has gained popularity is bamboo.

Bamboo is a highly sustainable resource that offers numerous advantages over traditional materials like wood and plastic. Its fast growth rate, low environmental impact, and versatility make it an ideal choice for a wide range of products.

In this article, we will explore the advantages of bamboo as a sustainable resource, its historical significance in Vietnam and Southeast Asia, the traditional uses of bamboo, the rise of bamboo products in the global market (<https://farmvina.com/rattan-bamboo-in-the-global-market/>), and the challenges and opportunities for bamboo producers and exporters



<https://farmvina.com/bamboo/>

How The Bamboo Products can improve ...? Discuss in Class?



<https://www.fao.org/forest-farm-facility/forest-farm-facilitynews/news-detail/ar/c/1734887/>

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Case study in Vietnam: Discussion Questions to Promote Bamboo Handicraft for Job Creation in Marginalized Communities:

1. How can local governments and NGOs support ethnic minority and mountainous communities in building sustainable bamboo-based craft businesses? (Explore training, funding, and market access solutions.)
2. What strategies can be implemented to connect traditional bamboo artisans with national and international markets to ensure fair income and employment stability?
3. How can digital tools and e-commerce platforms empower disadvantaged and rural communities to sell bamboo handicrafts more effectively?
4. What role should education and vocational training play in preserving traditional bamboo craftsmanship and adapting it for modern product design and global trends?
5. How can bamboo and rattan value chains be developed to ensure that all community members—including women, elderly, and low-skilled workers—benefit from employment opportunities?

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Case study: Kakadu Plum

The Kakadu plum is native to the top end of Northern Australia and is the richest known source of Vitamin C in the world, with multiple potential applications in the food, beauty and health industries.

The Gundjeihmi Aboriginal Corporation, representing the Mirarr people, says:

The Kakadu plum has been an important source of food and medicine for the Mirarr...It also features in oral histories and 'dreaming' stories.

Local Indigenous people are now beginning to trial commercial plantations of the fruit, however there were concerns in the past that Indigenous communities had little or no recognition, control or benefit-sharing from the commercialisation of their knowledge.

Wendy Morgan, chair of the Gandangara Local Aboriginal Land Council, said:

You'll have the big pharmaceutical companies coming out and talking to [Indigenous communities] and taking samples of their medicines. They might acknowledge where they got it from, but there is no money going back into that community they got the information from.

In a 2016 episode of Landline, Margo Northey, leader at the Wadeye Women's Centre noted:

This is knowledge that's been around for a long time, and it's great that it's being exploited to some extent, but recognition...is really important. There is no priority given to Indigenous people. Anyone, including big companies, can currently put in place patents on processing bush foods, making it difficult for Indigenous people to commercialise them.

The program also notes that 'the Northern Land Council...represents traditional owners and is calling for a blanket moratorium on all patents over native foods and plants until a legal framework protecting Indigenous interests can be enforced.'

Case study in Vietnam: The Emergence of Lotus Farming as an Innovation for Adapting to Climate Change in the Upper Vietnamese Mekong Delta

tourism based on visits to lotus fields.

Problem: The Upper Vietnamese Mekong Delta faces increasing challenges from climate change, including rising floods and saline intrusion, threatening traditional rice farming and livelihoods.

Innovation: Lotus farming is emerging as a resilient and profitable alternative to conventional rice cultivation. It offers:

- ❑ **Climate Adaptability:** Lotus thrives in flooded conditions and is more tolerant to water fluctuations than rice.
- ❑ **Economic Benefits:** Higher and more diverse income streams from various lotus products (seeds, flowers, stems, roots, leaves) compared to rice.
- ❑ **Resource Efficiency:** Potential for reduced input costs (e.g., less fertilizer, pesticide).



Article: <https://edepot.wur.nl/545949>

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How The Lotus Products can improve ...? Discuss in Class?



<https://en.vietnamplus.vn/do-ng-thap-diversifies-lotus-products-post304593.vnp>

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Case study in Vietnam: Discussion questions to encourage the development of industries and job creation related to lotus-based products for the benefit of ethnic minorities, ethnic groups in mountainous areas, and disadvantaged workers

1. How can local governments and development organizations support ethnic minority communities in accessing training and resources to cultivate and process lotus products?
2. What community-based business models (e.g., cooperatives, social enterprises) could be established to promote lotus farming and product development among disadvantaged labor groups?
3. How can cultural knowledge and traditional skills of ethnic minorities be integrated into the design and marketing of lotus-based handicrafts and wellness products?
4. What kinds of vocational training programs are needed to help women, youth, and ethnic minorities develop value-added lotus products for domestic and export markets?
5. How can ecotourism linked to lotus fields be leveraged to create inclusive job opportunities for local marginalized communities? · · ·
6. What incentives or micro-financing mechanisms could be introduced to encourage small-scale lotus entrepreneurs from underprivileged backgrounds? ·

